

Lab Report for Power Electronics

Lab 3: Charge pump and Step down

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1 INTRODUCTION

1.1 Research Question/ Problem Statement

Discover the working of some charge pump supplies and a step down converter

2 IC will be investigated:

- 1) LT1054
- 2) LM2575

1.2 Background/ Theory

A switched power supply uses energy storage elements like capacitors, inductors and/or transformers to store the energy.

2 METHOD & MATERIALS

2.1 Experimental Set-up

The given circuits are the ones from the lab text. We refer to the lab text for these circuits.

2.2 Measuring Instruments

This lab took place in the lab 10.03. The instruments used and their specifications are listed below.

- Aim TTI power supply EL302RD
- Rhode and Schwarz HMO 1002
- Tektronix 011-0106-00 current probe

Since the specifications of these devices are less relevant for these measurements we refer to the datasheets of these devices.

2.3 Method

By measuring voltage and current and relate this to the input signals, the behavior of the circuit can be determined and an explanation for the behavior can be formulated.

3 RESULTS VOLTAGE INVERTER

3.1 Measurements

One common application of charge pumps is the **voltage inverter** circuit, which converts a positive input voltage (V_{in}) into a corresponding negative output voltage (approximately $-V_{in}$).

For this experiment, we utilize the **LT1054** charge pump IC, which simplifies the construction of a voltage inverter. The LT1054 includes internal switching circuitry that automatically handles the charge transfer between capacitors, providing an efficient way to achieve the desired negative output voltage. By connecting external capacitors of appropriate values, we configure the LT1054 to generate a negative voltage from a positive input supply.



Figure 1: Voltage Inverter Input-Output

The inverter is expected to output approximately $-5V$ from a $+5V$ input. The deviation from this ideal output is likely due to losses in the components, namely the forward voltage drop of the diodes and internal resistance of the capacitors.

The internal resistance of the capacitors (ESR) causes power dissipation in the form of heat during the charge and discharge cycles. This can lead to a voltage drop and increase the ripple voltage.

The switching mechanism creates losses due to the finite time it takes for the switch to transition between on and off states.



Figure 2: Voltage Inverter Input Ripple



Figure 3: Voltage Inverter Output Ripple

The waveform shows a switching ripple pattern typical of a voltage inverter circuit. The ripple voltage amplitude is small but present, which is expected due to the charge and discharge cycles of the capacitors in the inverter circuit. If we want to reduce the ripple further, we could increase the value of the output capacitor (C_2) or use capacitors with lower ESR.

3.2 Conclusion

We successfully built and tested a voltage inverter circuit to convert a 5V input to a $-5V$ output using a charge pump configuration. The output was measured with an oscilloscope, showing a small ripple in the output voltage, which is typical of such circuits due to the switching operation. The output was lower than 5V due to the before mentioned losses.

Overall, the experiment demonstrates the fundamental operation of a charge pump inverter, and the results are within expected ranges. To improve performance, the ripple could be reduced by using lower ESR capacitors and increasing the capacitor values for better filtering. This experiment provided a solid understanding of voltage inversion in DC-DC converters and how to optimize performance.

4 RESULTS VOLTAGE DOUBLER

4.1 Measurements

In this experiment, we explore the operation of a **charge pump voltage doubler**, a type of DC-DC converter that can double the input voltage without requiring inductive components. The **voltage doubler** works by utilizing two capacitors, C_1 and C_2 , along with switching elements to sequentially transfer and stack voltages. For this experiment, we utilize the **LT1054** charge pump IC, which simplifies the design and operation of a voltage doubler by incorporating the necessary switching circuitry.

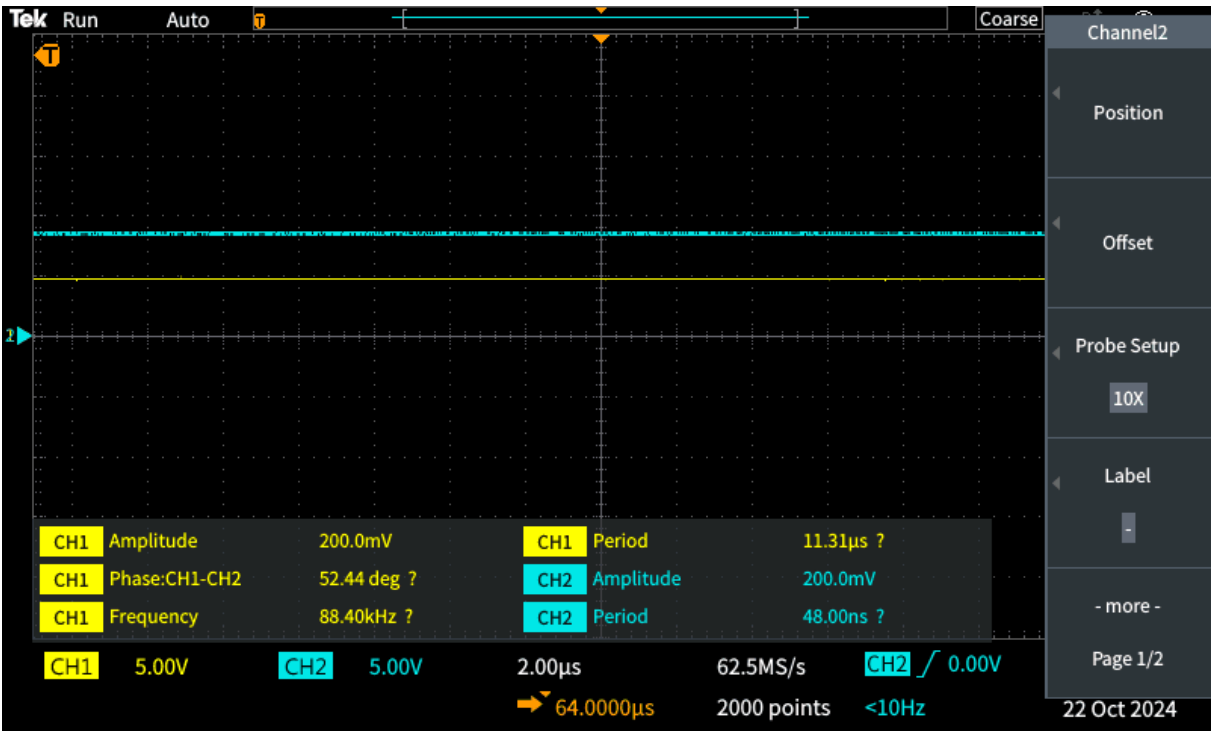


Figure 4: Voltage Doubler 5V Input

The voltage doubler circuit theoretically outputs twice the input voltage, but there are losses due to capacitor ESR, parasitic resistances, and the switching mechanism. This is likely why the output voltage is not exactly twice the input voltage

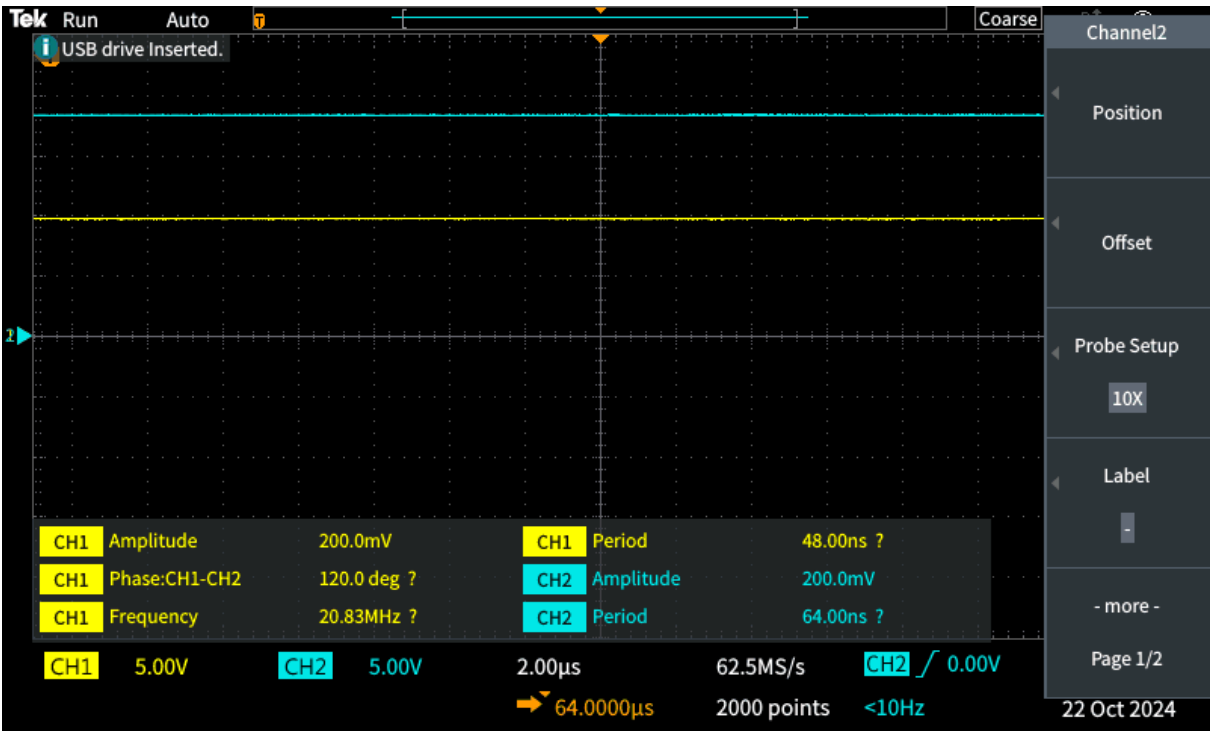


Figure 5: Voltage Doubler 10V Input

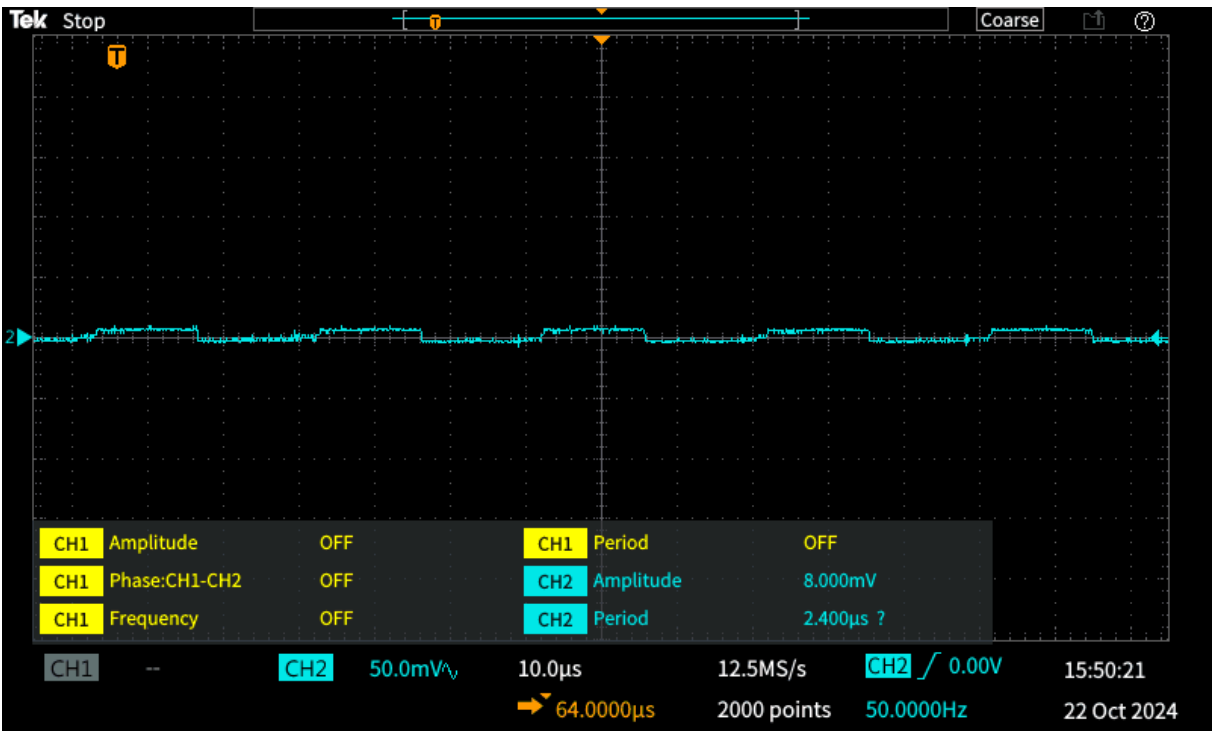


Figure 6: Voltage Doubler Output Ripple

The small output ripple observed is unlikely to have a significant impact on the performance of the voltage doubler, as it is well within acceptable limits for most low-power or low-sensitivity

applications. The output ripple is minimal, indicating that the voltage doubler is functioning efficiently and providing a clean,

4.2 Conclusion

The circuit exhibited stable performance, with a clean and well-regulated output voltage. The low ripple amplitude confirmed that the voltage doubler was operating efficiently with minimal fluctuation, making it suitable for applications requiring a stable DC voltage

5 RESULTS STEP DOWN

5.1 Measurements

The final circuit we will build is the step-down converter using the LM2575 in figure 7 below. This circuit will be used to make a 9V power supply using an input voltage of $V_{in} = 15V$ and with a maximum output current of $I_{out} = 0.8A$. The component values for R_1, R_2, L_1, C_{out} are calculated based on the datasheet and are: $R_1 = 3.48 \approx 3.3k\Omega, R_2 = 22k\Omega, L_1 = 330\mu H, C_{out} = 100\mu F$.

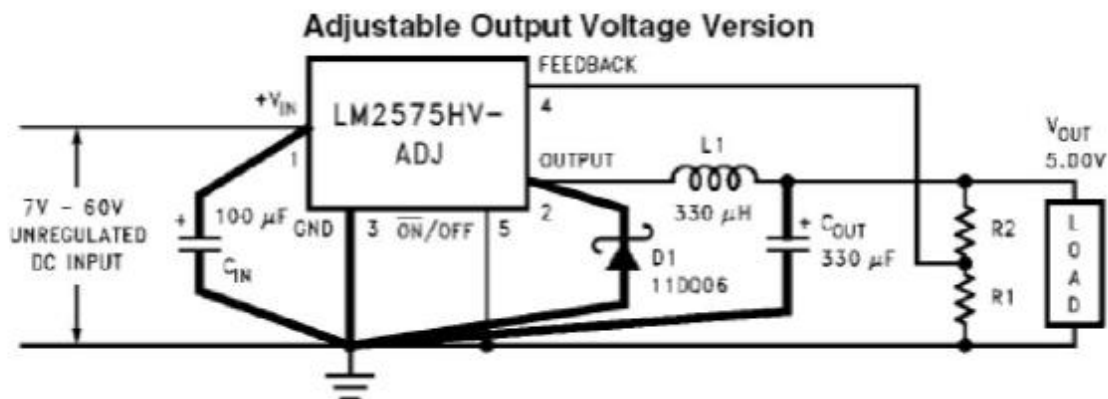


Figure 7: Step-Down Converter Circuit (LM2575)

We first build this circuit and measure the output voltage when using a 470Ω resistor as the load. Figure 8 below shows the output voltage as a function of time when using an input voltage signal of 15V.

5.2 Graphs

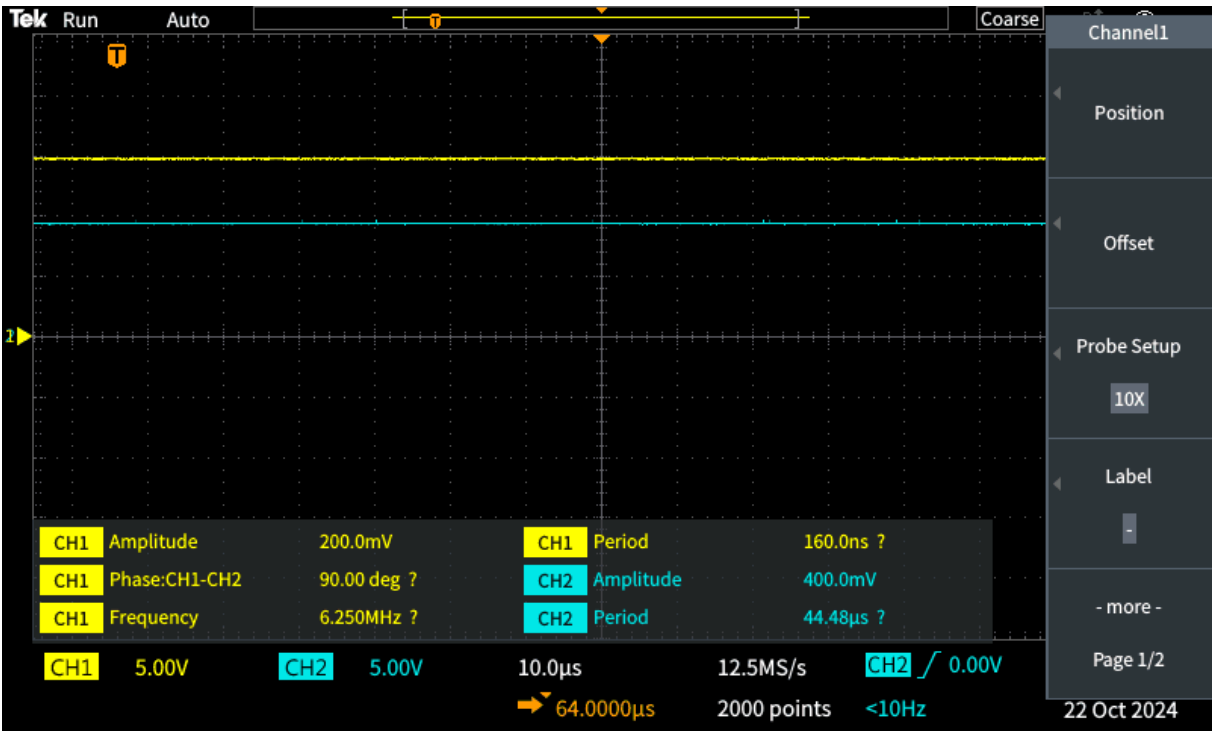


Figure 8: Step-Down Converter 15V Input

As we can see, the circuit correctly steps down the input voltage from 15V down to 9V at the output. We then measure the voltage and current at pin 2 of the converter.

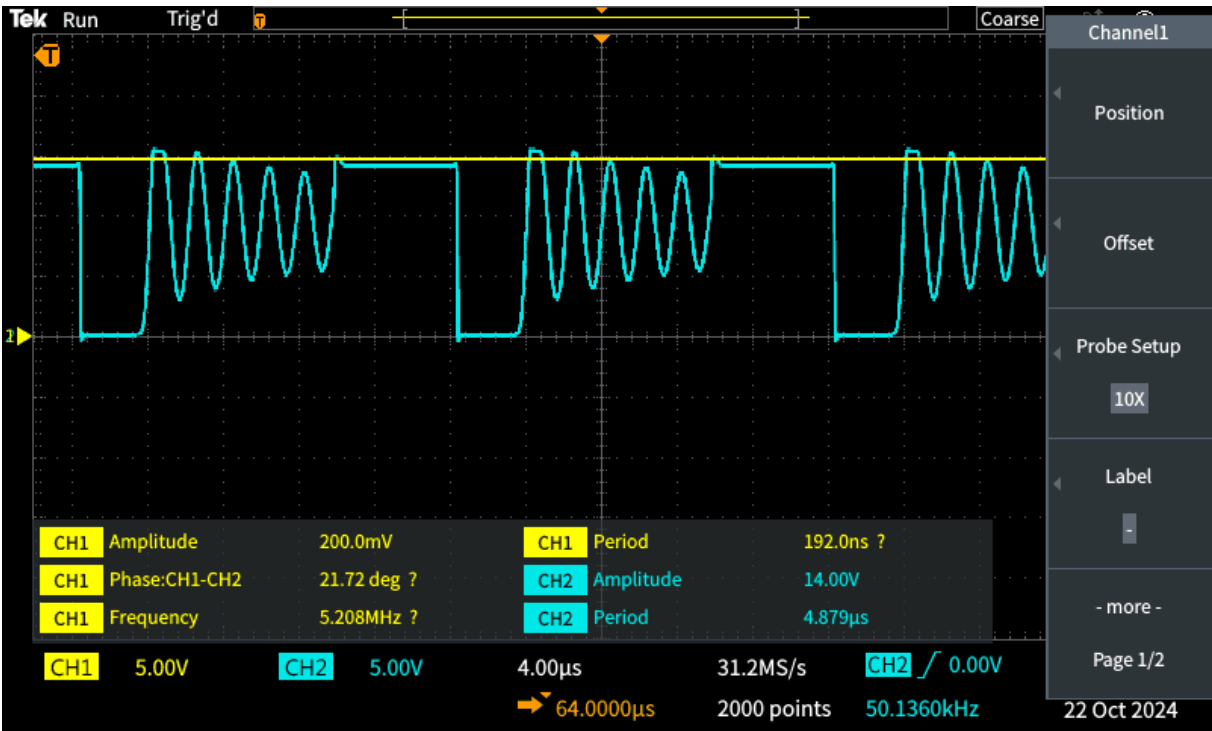


Figure 9: Step-Down Converter Pin 2 Signal 470Ω



Figure 10: Step-Down Converter Current Signal 470Ω

The figure above shows the voltage (CH1) and the current (CH2) at pin 2 of the converter. From the data sheet of the LM2575, we know the component has a 52 kHz (typical) fixed-frequency internal oscillator. Looking at current at pin 2, we see the signal has a period of about 20μs which corresponds to a frequency of 50 kHz as expected. Additionally, we note that the LM2575 is operating in discontinuous mode given that the inductor current is not flowing continuously but rather goes to zero during the normal switching cycle. As expected, any buck regulator eventually operates in discontinuous mode when the load current is light enough. Moreover, we notice that at the moment of switching i.e. voltage goes from high to low, the inductor begins to discharge as can be noted from the current waveform decreasing and eventually reaching zero upon switching.

We then repeat the measurement but using a 24Ω load this time in place of the 470Ω load. The main difference is that our voltage regulator now operates in continuous mode as can be seen in the figures below. The current never settles to zero, and instead continuously flows through the inductor.

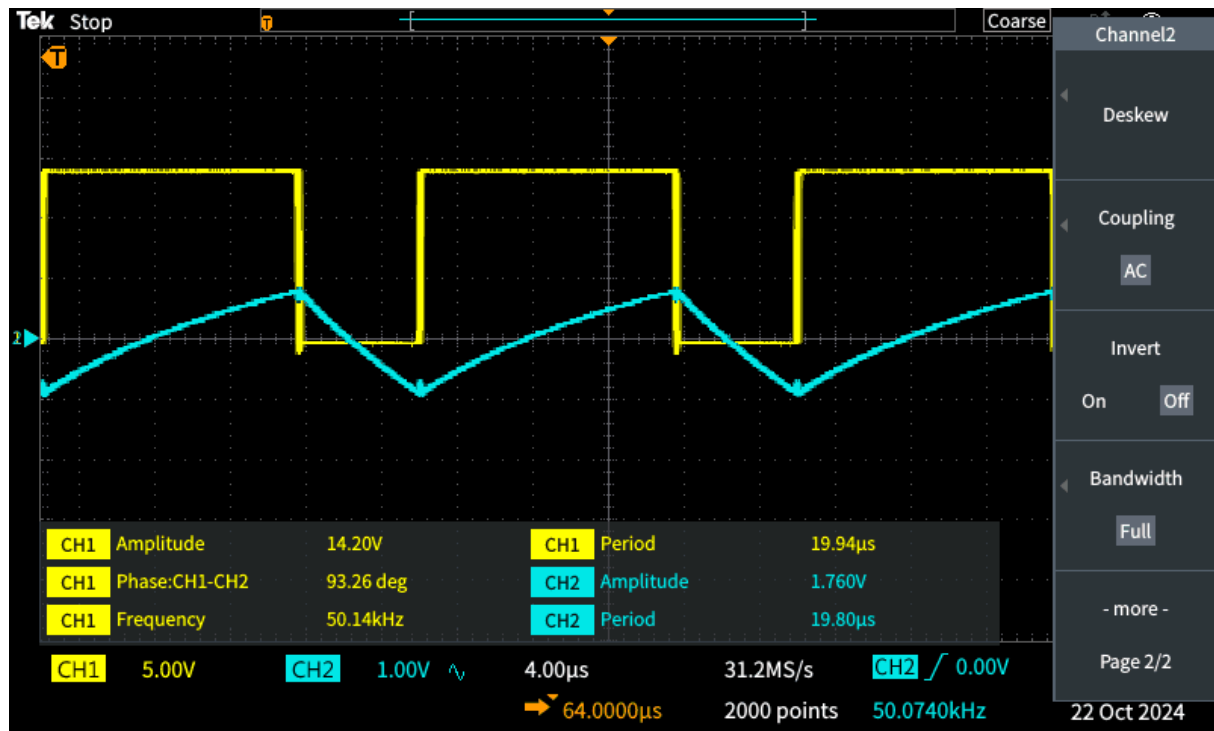


Figure 11: Step-Down Converter Current Signal 24Ω

We observe a smoother current due to the continuous mode of operation and also notice that any oscillatory behaviour due to discontinuous mode is no longer present. Nonetheless, we note the frequency of the signal remains the same as before (50.14 kHz) which is expected given the fixed nature of the internal oscillator. Similarly to the previous figures, we see as soon as the voltage switches from high to low, the current begins to decrease. In fact, we notice that the general characteristics of this mode of operation is similar to the discontinuous mode in the sense that the current still exhibits similar behaviour but simply that it is continuously flowing rather than settling to zero. We do notice however that the ripple on the voltage at pin 2 using the 24Ω resistor is removed in contrast to when using the 470Ω resistor. The efficiency of our circuit is 80% which remains relatively high but is still subject to power losses. The inductor, L_1 , in this circuit is not a perfect inductor meaning it has an ESR in which power is dissipated leading to lower efficiency. A similar reasoning can be applied to capacitor, C_{out} . The parasitic characteristics of the components used in our circuit will lead to power losses. Additionally, the layout used to build this circuit, i.e. on a breadboard is not ideal which will give rise to noise and oscillations.

6 GENERAL DISCUSSION

The main objective of the experiment was to explore the functionality of charge pump circuits: a voltage inverter, a voltage doubler, and a step-down and to assess their ability to achieve the expected output voltages. Each setup was tested to determine if the output aligned with theoretical predictions, and measurements were taken to validate circuit performance.

The results were consistent with theoretical expectations: the **voltage inverter** produced an output close to $-V_{in}$, while the **voltage doubler** provided approximately $2V_{IN}$. The measurements aligned with the expected values within the margin of error, confirming that the circuits functioned correctly under ideal conditions. Minor deviations in the output could be attributed to parasitic losses in components, which were within acceptable ranges for this lab. Similarly, the **step-down** converter, as expected, converted the 15V input signal down to a stable 9V signal at the output.

7 GENERAL CONCLUSION

The **voltage inverter** effectively converts a positive input voltage to a corresponding negative output voltage, demonstrating the ability of charge pumps to generate opposite polarity without inductive components. The **voltage doubler** successfully boosts the input voltage to approximately twice its original value, illustrating the efficiency of charge pump circuits in achieving higher voltages in low-power applications. The **step-down** (buck) converter is an effective DC-to-DC converter which decreases voltage, while increasing current, from its input supply to its output load. It is a simple and effective class of switched-mode power supply.